Quantum-Inspired Cognitive Networking: A Framework for Adaptive AI-Native 6G Infrastructure

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ABSTRACT

AI-native 6G networks require routing policies that can expose alternative paths under changing conditions. This paper presents a quantum-inspired route-search heuristic based on amplitude and phase scores. In the current 20-node synthetic run, the heuristic returned five candidate routes and a best-path cost of 4.0, while the classical Dijkstra baseline returned one path with cost 2.0. Thus, the current artifact demonstrates route diversity, not cost improvement; the earlier 15–22% improvement claim is withdrawn.

KEYWORDS

quantum-inspired, cognitive networking, 6G, adaptive routing, AI-native

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1 INTRODUCTION

Classical routing algorithms optimize a scalar path objective. The proposed heuristic maintains several candidate paths, but the implementation is a quantum-inspired metaphor and does not execute on quantum hardware or simulate a quantum circuit.

2 RELATED WORK

Quantum-inspired evolutionary algorithms represent candidate solutions with amplitudes or phase-like parameters and update them using classical arithmetic. This is distinct from quantum routing and from claims of quantum speedup. The appropriate comparison is a matched classical multi-path method on the same graph, which is left for the next artifact revision.

3 METHODOLOGY

3.1 Superposition Exploration

Candidate selection probability:

$$P(j|i) = |A_{ij}|^2 \cdot (1 + \cos(\phi_{ij})) \tag{1}$$

where A_{ij} is the amplitude and ϕ_{ij} is the phase between nodes i and j.

4 EXPERIMENTAL PROTOCOL

The seeded experiment uses 20 nodes, source 0, destination 19, five candidate routes, and NumPy seed 20260909. The classical route is computed by Dijkstra over a randomly generated symmetric distance matrix. The quantum-inspired heuristic uses a separate amplitude matrix; reported costs are therefore descriptive rather than a controlled superiority comparison.

4.1 Entanglement Correlation

$$E(r_1, r_2) = \frac{|r_1 \cap r_2|}{\max(|r_1|, |r_2|)} \tag{2}$$

5 RESULTS

Table 1: Route Cost Comparison (20 nodes)

Method	Cost	Routes Found
Quantum-Inspired	20.0	5
Classical Dijkstra	2.0	1

The quantum-inspired optimizer discovered five candidate routes. The best candidate cost was 20.0, compared with 2.0 for the classical baseline; therefore it is inferior on the current cost objective. Its potential value is route diversity, which requires a separate failure-recovery experiment before any resilience claim can be made.

6 LIMITATIONS AND THREATS TO VALIDITY

The current graph is randomly generated, the optimizer does not enforce a physical neighbor topology, and no repeated-seed confidence intervals or link-failure trials are included. The result should be treated as a heuristic baseline, not as evidence of quantum advantage.

7 CONCLUSION

The artifact demonstrates candidate-route generation but does not demonstrate lower routing cost. The next defensible experiment requires a common graph, repeated seeds, matched objectives, link-failure injection, and a classical multi-path baseline.

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